

YU FAN

Male | Age 44 | +86 13918677360 | yuff100@163.com
19 Years of Experience | Technical Director | Preferred Location: Shanghai

PERSONAL STRENGTHS

- Open to work location; business travel acceptable
- 20 years of high-availability, high-performance backend R&D
- 10+ years managing teams of 30+ engineers
- 4 years of overseas work experience
- Tech stack: C / C++ / Go / Python
- Server-side architect, DevOps architect, innovation prototyping
- Carrier core network, 4G/5G RAN, NPN private network solutions
- VPP/DPDK high-performance media server architecture
- Cloud-native architecture, CI/CD
- Entrepreneurial mindset for business growth and team building
- VIA Strengths: Teamwork, Multi-perspective, Fairness, Humility, Excellence
- MBTI: ENFJ (Extraversion, Intuition, Feeling, Judging)
- Personal website: www.deepnomind.com
- InfoQ Featured Creator – 2021 & 2022

WORK EXPERIENCE

Mavenir Systems R&D Director May 2008 – Present

Responsibilities:

1. Team building and performance management.
2. Project planning and scheduling.
3. Business requirements delivery and architecture design.
4. Participated in R&D of multiple IMS core network elements including P-CSCF, IBCF, ATCF, E-CSCF, I-CSCF, S-CSCF, TAS. Tech stack: Linux / C / C++.
5. Led media gateway middleware development providing media services for multiple products including WebRTC and SBC, supporting VoLTE/VoWiFi solutions. Tech stack: Linux / C / C++ / DPDK / VPP.
6. Designed Layer-7 load balancing architecture supporting NFV evolution. Tech stack: Linux / C / C++.
7. Designed and implemented 4G/5G RAN CU; mentored team through cloud-native architecture evolution. Tech stack: Linux / C / Go.
8. Built R&D CI/CD team, defined processes, and developed automated test and deployment scripts. Tech stack: Linux / Python / Jenkins.
9. Built 4G/5G RAN architect team and tracked emerging technologies.
10. Managed NPN private network automated deployment team and business integration. Tech stack: Linux / Python / Java.
11. Innovation prototyping and architecture delivery (RAN DU uBPF integration, AI real-time voice translation).

Achievements:

1. Completed 5G CUUP microservice transformation.
2. Completed separation of 4G CU control plane and user plane.
3. Built RU PnP OAM microservice architecture supporting dynamic automated onboarding and configuration of 8,000+ RUs.
4. Introduced VPP/DPDK to accelerate user-plane processing for SBC and 4G/5G CUUP high-speed data.
5. Planned and built 10+ CI/CD pipelines empowering multiple project R&D and QA teams.
6. Built a proprietary automated testing framework supporting 4G CU, 5G CUUP, SDaaS, IMS, SBC and other projects.
7. Built an AI real-time voice translation server on the UAG platform supporting both cloud inference and on-device inference; deployed to T-Mobile US live network.

Motorola Software Engineer May 2007 – May 2008

Responsibilities:

- Developed iDEN handset applications.
- Developed automated test scripts based on STRIDE and designed test cases.
- Developed PoC (Push-to-talk over Cellular) application server.

Achievements:

1. Designed and implemented the PTT server.
2. Designed test cases for multiple handset features.
3. Provided automated testing framework support for multiple handset features.

PROJECT EXPERIENCE

AI Voice Application Service Architect Jul 2025 – Present

Designed the architecture of a real-time voice translation service and seamlessly integrated it into an existing carrier core network, enabling real-time AI voice translation capabilities. Led a 5-person team for technology selection, prototype design, and architecture. Achieved efficient integration and optimization of the voice translation service, improving user experience and translation accuracy. The solution supports both on-device inference and third-party cloud services.

Radio Access Network Group Director Oct 2019 – Jun 2025

1. Led R&D of 4G/5G base station products, ensuring technology innovation aligned with industry standards.
2. Collaborated with the team to implement base station functions on cloud-native architecture, improving system performance and reliability.
3. Led foundational infrastructure setup to ensure the technical framework met long-term development needs.
4. Proactively communicated with business teams to deeply understand requirements and convert them into concrete technical solutions.
5. Supervised project progress and ensured on-time delivery of high-quality results through effective project management.
6. Regularly assessed system architecture and continuously optimized it to meet rapidly changing market demands.

Private Wireless Network Solution R&D Director Apr 2022 – Dec 2022

1. Led R&D of the Non-Public Network (NPN) project, ensuring project goals aligned with company strategy.
2. Responsible for team building and management, optimizing team structure and overall effectiveness.
3. Closely aligned with business teams to ensure accurate understanding and implementation of project requirements.
4. Formulated and executed project plans; monitored project progress to ensure on-time delivery.
5. Drove R&D of automated deployment solutions, improving deployment efficiency and system stability.
6. Managed Web GUI development, ensuring user-friendly and functional interfaces.
7. Implemented O-RAN/5GC integration to enhance network performance and service quality.
8. Regularly organized project reviews and proposed solutions for technical challenges and schedule issues.

Unified Access Gateway R&D Manager Jan 2013 – Oct 2019

Led a team of 12 engineers in media server R&D. Supported development of RTP, MSRP, IPSec, H.248 protocols and related services. Participated in all stages including product requirements, architecture design, module implementation, unit testing, integration testing, system testing, and field support. Customers included T-Mobile, DTAG, Reliance, Tele2, AMX and other major global carriers.

IMS Core Network Technical Lead May 2008 – Jan 2013

Participated in R&D of multiple company products including I-CSCF, S-CSCF, IP-SM-GW, RCS, SMSC, MMSC. Designed and developed SIP, CDR, Call Control and various other modules. Customers included T-Mobile, DTAG, Vietnam Telecom and other telecommunications operators.

Motorola Handsets Software Engineer Apr 2007 – May 2008

- Developed iDEN handset applications.
- Developed automated test scripts based on STRIDE and designed test cases.
- Developed PoC (Push-to-talk over Cellular) application server.

EDUCATION

Nanjing University of Posts and Telecommunications Master's Degree | Computer Application Technology 2004–2007

Nanjing Normal University Bachelor's Degree | Electronic Information Engineering 2000–2004

CERTIFICATIONS & QUALIFICATIONS

College English Test Band 6 (CET-6) | College English Test Band 4 (CET-4) | National Computer Level 3
Coursera Certified: Soft Skills | Google Certified: Project Management Fundamentals
Coursera Certified: Communication Skills | Coursera Certified: Time & Stress Management | Coursera
Certified: Conflict Management